

Hospital AI Agent

MVP Development Plan

1. Introduction

The Hospital AI Agent project contains many advanced features related to hospital management, artificial intelligence, appointments, notifications, and healthcare support.

To make the development process organized and practical, the project will be developed in phases.

The first phase will focus on building a Minimum Viable Product (MVP).

The MVP is the first working version of the system. It includes the most important features required to demonstrate the main idea of the project.

2. MVP Goal

The main goal of the MVP is to create a working hospital management system with a basic AI Agent.

The first version should allow users to:

- Manage doctors.
- Manage patients.
- Book appointments.
- View appointment information.
- View basic hospital statistics.
- Interact with an AI Agent.
- Search for doctors.
- Check appointment availability.
- Receive AI-assisted appointment recommendations.

The MVP will focus on core functionality before implementing advanced features.

3. MVP Scope

The first version of the project will include the following components.

User Management

The system will support basic user authentication and authorization.

Users included in the MVP:

- Administrator.
- Doctor.
- Patient.

Additional roles such as Nurse and Receptionist may be added in later phases.

4. Phase 1: Core System Development

The first phase focuses on building the main hospital management system.

4.1 Authentication

The system will provide:

- User registration.
- User login.
- Secure password storage.
- JWT authentication.
- Role-based access control.

Supported roles:

- Admin.
 - Doctor.
 - Patient.
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4.2 Database Setup

The initial database will include the following tables:

- Users.
- Departments.
- Doctors.
- Patients.
- Appointments.

The database will be implemented using PostgreSQL.

4.3 Doctor Management

The Administrator will be able to:

- Add doctors.
- View doctors.
- Edit doctor information.
- Delete doctors.
- Assign doctors to departments.
- Define doctor working hours.

Each doctor will have:

- Name.
 - Specialization.
 - Department.
 - Working hours.
 - Consultation fee.
 - Availability status.
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4.4 Patient Management

The system will allow authorized users to:

- Register patients.
- View patient information.
- Search for patients.
- Update patient information.

The system will initially store basic patient information.

4.5 Appointment Management

The appointment system is one of the main features of the MVP.

Users will be able to:

- View doctors.
- Select a specialty.
- Select a doctor.
- Check available appointment slots.
- Book an appointment.
- Modify an appointment.
- Cancel an appointment.
- View appointment status.

Appointment statuses will include:

- Booked.
 - Completed.
 - Cancelled.
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5. Phase 2: Dashboard Development

The second phase focuses on the administration dashboard.

The dashboard will display:

- Total number of doctors.

- Total number of patients.
- Total appointments.
- Daily appointments.
- Completed appointments.
- Cancelled appointments.

The dashboard may also include:

- Charts.
 - Appointment statistics.
 - Department statistics.
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6. Phase 3: AI Agent MVP

This phase introduces the first version of the AI Agent.

The AI Agent will be able to understand basic user requests.

Example:

“I want to book an appointment.”

The Agent will:

1. Understand the user request.
 2. Identify the user's intention.
 3. Determine the required information.
 4. Select the appropriate tool.
 5. Retrieve information from the backend.
 6. Provide a useful response.
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6.1 AI Agent Feature: Search Doctor

Example request:

“I need a heart doctor.”

The AI Agent will:

1. Identify the requested specialty.
 2. Search the Doctors table.
 3. Find doctors related to Cardiology.
 4. Display available doctors.
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6.2 AI Agent Feature: Check Availability

Example request:

“When is the earliest available appointment?”

The AI Agent will:

1. Identify the selected doctor or specialty.
 2. Check doctor schedules.
 3. Check existing appointments.
 4. Find available appointment slots.
 5. Suggest the best available options.
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6.3 AI Agent Feature: Appointment Assistance

Example request:

“Book the earliest appointment.”

The AI Agent will:

1. Confirm the required doctor or specialty.
2. Search for available appointments.
3. Present available options.
4. Receive user confirmation.
5. Call the appointment booking tool.
6. Confirm the booking.

6.4 AI Agent Feature: Hospital Questions

The AI Agent will answer general hospital questions.

Examples:

- What doctors are available today?
- What specialties are available?
- When is my appointment?
- Which doctor is available for a specific specialty?

7. MVP AI Agent Tools

The first version of the AI Agent will use controlled tools.

Tool 1: Search Doctor

Purpose:

Search for doctors based on:

- Specialization.
- Department.
- Doctor name.

Tool 2: Check Availability

Purpose:

Check available appointment slots for a doctor.

Tool 3: Book Appointment

Purpose:

Create a new appointment after receiving the required confirmation.

Tool 4: Get Appointment Information

Purpose:

Retrieve appointment details for an authorized user.

8. MVP User Flow

The basic patient flow will be:

Patient

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Login

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View Doctors

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Select Specialty

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Select Doctor

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Check Availability

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Book Appointment

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Receive Confirmation

The patient can also interact with the AI Agent during the process.

9. MVP AI Agent Flow

The AI Agent workflow will be:

User Request

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Understand Request

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Identify Intent

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Select Tool

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Call Backend API

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Retrieve Database Information

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Process Result

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Provide Response

10. Technology Stack for MVP

Frontend

React.js

Backend

Python + FastAPI

Database

PostgreSQL

AI Agent

Python with Large Language Model integration.

Authentication

JWT

API

REST API using FastAPI.

11. MVP Development Steps

The development process should follow this order.

Step 1

Create the project repository and development environment.

Step 2

Design and create the PostgreSQL database.

Step 3

Develop the backend using FastAPI.

Create APIs for:

- Authentication.
 - Doctors.
 - Patients.
 - Appointments.
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Step 4

Test backend APIs.

Step 5

Develop the frontend using React.

Create:

- Login page.
- Dashboard.
- Doctor pages.
- Patient pages.
- Appointment pages.

Step 6

Connect the frontend with the backend.

Step 7

Implement the AI Agent.

Start with:

- Search Doctor.
- Check Availability.
- Appointment Assistance.

Step 8

Connect the AI Agent with backend tools.

Step 9

Test the complete system.

12. MVP Deliverables

At the end of the MVP phase, the team should have:

- A working web application.
 - User authentication.
 - Role-based access.
 - Doctor management.
 - Patient management.
 - Appointment management.
 - Basic dashboard.
 - PostgreSQL database.
 - FastAPI backend.
 - React frontend.
 - Basic AI Agent.
 - AI Agent tools.
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13. Phase 2 Development

After completing the MVP, additional features will be implemented.

These include:

- Nurse management.
- Receptionist management.
- Nurse shifts.
- Notifications.
- Email reminders.
- Prescription management.
- Medication reminders.
- Advanced reports.

14. Phase 3: Advanced AI Features

The final phase will focus on advanced artificial intelligence features.

These may include:

- Workload analysis.
 - Hospital crowd prediction.
 - Smart nurse distribution.
 - Intelligent report generation.
 - Patient history summarization.
 - Advanced appointment recommendations.
 - Voice interaction.
 - Multi-language support.
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15. Development Priority

The project development priority is:

Priority 1:

Authentication

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Priority 2:

Database

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Priority 3:

Doctor and Patient Management

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Priority 4:

Appointment Management

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Priority 5:

Dashboard

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Priority 6:

AI Agent

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Priority 7:

Notifications

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Priority 8:

Advanced AI Features

16. Success Criteria

The MVP will be considered successful if:

- Users can log in successfully.
 - Administrators can manage doctors.
 - Patients can be registered and managed.
 - Appointments can be booked successfully.
 - Appointment conflicts are prevented.
 - The dashboard displays basic statistics.
 - The AI Agent can answer basic hospital questions.
 - The AI Agent can search for doctors.
 - The AI Agent can check appointment availability.
 - The AI Agent can assist with appointment booking.
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17. Conclusion

The MVP Development Plan provides a clear roadmap for building the Hospital AI Agent project.

The project will begin with the essential hospital management functions, including users, doctors, patients, and appointments.

After the core system is stable, the AI Agent will be integrated with controlled tools to provide intelligent assistance.

Advanced hospital management and AI features will be added gradually in future development phases.

This phased development approach will make the project easier to build, test, maintain, and demonstrate.